

Natural Resources and Their Use

When does Nature become a resource? How do we sort resources, why is their uneven spread a source of both wealth and war — and how do we use them without using them up?

The one-line story of this chapter

Nature exists on its own. The moment humans take something from Nature to use it, that thing becomes a **resource**. But not everything in Nature is a resource — it must be **reachable, affordable and acceptable** to use. Because resources are spread **unevenly** across the Earth, they shape where people live, how they trade, and even why they fight. The chapter ends with a plea: use renewables gently so they regenerate, and use non-renewables wisely so they last — this is **stewardship**.

When does Nature become a "resource"?

Trees, water, air, coal — these are all part of **Nature** and exist without humans. They become **resources** only when humans use them for survival or to make something. A tree is Nature; the wooden chair you make from it treats the tree as a resource.

The 3 tests for a resource (very high-yield): for something in Nature to count as a resource it must be —

- **Technologically accessible** — we have the technology to reach/extract it.
- **Economically feasible** — the cost of extracting it is worth it.
- **Culturally acceptable** — society allows its use (e.g. cutting trees in a *sacred grove* is not acceptable).

Petroleum deep under the ocean that we cannot yet reach is *not* a resource — until it passes all three tests.

Exam word: here "**exploitation**" is *not* negative — it simply means the **extraction, utilisation and consumption** of natural resources.

Def → **Natural resources** are materials and substances that occur in Nature and are valuable to humans.

Def → **Utility** (usefulness) is what makes an object or substance a resource. No utility = not a resource.

Two ways to categorise natural resources

We categorise things using shared **criteria** so we can talk about them with one word. Natural resources are sorted in two important ways:

A. By the use we put them to

Category	What it does	Examples
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Resources essential for life	Keep us alive; we cannot manufacture them	Air, water, food (soil)
Resources for materials	Turned into useful or beautiful objects	Wood, marble, coal, gold
Resources for energy	Power buildings, transport, production	Coal, water, petroleum, gas, sunlight, wind

B. By whether they renew: Renewable vs Non-renewable

	Renewable	Non-renewable
Meaning	Nature restores/regenerates them within our timescale	Formed over millions of years; can't be replaced at the rate we use them
Condition	Renewable only if we don't disturb Nature's rhythm (don't harvest faster than it regrows)	Will run out — must be used judiciously
Examples	Solar, wind, flowing water, timber (if regrown)	Coal, petroleum, iron, copper, gold

Common trap: A renewable resource can **become** non-renewable! If we cut timber faster than the forest regrows, or over-fish during spawning season, the resource is depleted and stops renewing. Renewability is a *condition*, not a permanent label.

Restoration vs Regeneration: **Restoration** = returning something damaged to its original healthy state (a cut heals, a forest recovers after fire). **Regeneration** goes further = Nature creates *new* life and conditions for thriving. Nature works in cycles with **no waste**.

Two "bonus" terms examiners love

Stock → **Stock of resources** = the **total available** amount of a resource (whether in use or not), i.e. the whole reserve.

Patent → A **patent** is the **exclusive right over any idea or invention** — no one else can use it without permission.

Ecosystem functions vs services: A forest naturally makes oxygen, filters water, prevents soil erosion, gives habitat — these are **ecosystem functions**. When *humans benefit* from them (clean water, protected farmland, pollinated crops), they are **ecosystem services**.

Memorable number: one mature tree makes about **275 litres of oxygen per day**, but a human needs about **350 litres per day** — so one tree isn't even enough for one person!

Uneven distribution — wealth, trade... and conflict

Natural resources are **not spread evenly** across the planet or within a country. This shapes settlements, trade, international relations and even wars.

- **Benefits:** industries near resources create jobs, townships grow, quality of life improves.
- **Costs:** people can be **displaced** from their homes; sacred places get threatened, causing conflict.
- **Shared waters cause tension:** Nature ignores political borders. E.g. the **Kaveri river** is shared (and disputed) by Karnataka, Tamil Nadu, Kerala and Puducherry; the **Brahmaputra** is shared across countries.

The "Natural Resource Curse" (paradox of plenty): having lots of natural resources does **not** guarantee riches. Countries that can't build industries to convert resources into higher-value products may stay poor. **India has largely avoided this curse** by investing in industries, technology and skills.

Sustainable Development & Stewardship — the heart of the chapter

Sustainable development means meeting our needs today **without** making it harder for **future generations** to meet theirs. Irresponsible use has caused **pollution, biodiversity loss and climate change**.

Principles of Sustainable Development (high-yield 3-marker):

- **Restore & regenerate renewables** — don't use them faster than they recover (e.g. recharge groundwater, rejuvenate soil).
- **Use non-renewables judiciously** — stretch them until sustainable alternatives are ready; switch to renewable energy.
- **Protect equity** — resources like clean water and air must reach *all* sections of society fairly.

Def → **Biodiversity loss** = the decline in the variety of life on Earth.

Idea → **Stewardship** = acting as careful caretakers of Nature. The Bhagavad Gītā's *lokasangraha* — act for the wellbeing of all, not just personal desire.

Case studies you must be able to name

Case	What happened	Lesson
Punjab groundwater	Green Revolution + free power + water-hungry HYV wheat/paddy → ~ 80% of Punjab now "over-exploited"; water below 30 m; chemicals in water	Short-term food security, long-term crisis
Cement	One of the most polluting industries (dust harms lungs, leaves, soil, water); CPCB sets guidelines; alternatives = mud, stone, plant & recycled materials	Traditional + modern = sustainable materials

Sikkim	Pema's farm switched to organic; in 2016 Sikkim became India's 1st 100% organic state ; incomes rose ~20%	A whole region can go sustainable
Bhadla Solar Park (Rajasthan)	One of the world's largest solar parks; powers ~15% of Rajasthan's needs	Shift to renewable energy
Vṛikṣhāyurveda	Ancient Indian botanical science (Surapala, ~10th c. CE) on plant care, soil, pests	India's traditional ecological wisdom

International Solar Alliance — launched by **India and France in 2015**, a coalition of sunshine-rich countries. Shows India's leadership in renewable energy.

Model answers — write them like this

Example A. "What can make what is today a renewable resource non-renewable tomorrow?" [3]

- Point 1:** A resource is renewable only while its natural rhythm of restoration and regeneration is not disturbed.
- Point 2:** If humans use it faster than it recovers — e.g. cutting timber faster than the forest regrows, or over-fishing tuna during spawning season — the stock collapses.
- Point 3:** Prevention: harvest within limits, allow regeneration, replant, and regulate use (like the old ban on fishing during spawning).

∴ Over-use beyond Nature's recovery rate turns a renewable resource into a depleted, non-renewable one.

Example B. "Explain the principles of Sustainable Development." [3]

- Idea:** Meet present needs without harming future generations' ability to meet theirs.
- P1:** Restore & regenerate renewables — don't exceed Nature's recovery rate.
- P2:** Use non-renewables judiciously so they last until greener options arrive.
- P3:** Ensure fair access to basic resources for all sections of society.

∴ Sustainable development = balance of ecology, economy and equity — stewardship of Nature.

Example C. "What is Stock of Resources?" [1]

Answer: The **total available** quantity of a resource — the entire reserve, whether currently in use or not.

∴ Stock = total available resources.

Don't lose easy marks

Iron ore is NON-renewable (a mineral/metal ore), not renewable. Coal & petroleum are non-renewable fossil fuels.

Utility (not beauty, quantity or availability) is what *makes* something a resource.

- Coal reserves in India may last only ~**50 more years** → use judiciously.
- Renewable examples: solar, wind, flowing water, timber (if regrown). Non-renewable: coal, oil, iron, copper, gold.
- Resource curse = "paradox of plenty"; India avoided it via industry + skills.

→ Now test yourself on the Worksheet, then check the Answer key.